5.4 Solving Special Systems of Linear Equations

Name _____ Period ____ Date _____

Algebra I 5.4 · Student sheet

I can…

- tell how many solutions a system has.
- solve systems with no solution or infinitely many solutions.
- use special systems to solve real-life problems.

Vocabulary · fill in the missing word

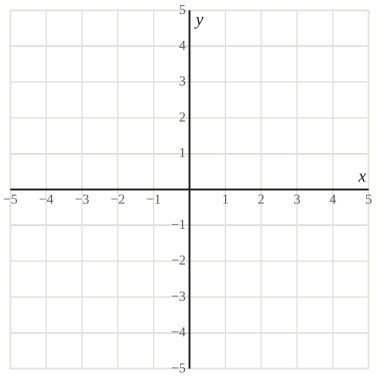
No solution — _____ lines (same slope, different y-intercepts); solving gives a false statement such as 0 = 5

Infinitely many solutions — the same line (same slope and _____); solving gives a true statement such as 0 = 0

One solution — _____ slopes

Part 1 · Three pictures

System	Graph (cross, parallel, or same line)	What happened when you solved?	Solutions
y = x + 1; y = −x + 3			
y = 2x + 1; y = 2x − 2			
y = x − 2; 2x − 2y = 4			



Part 2 · Notes and examples (fill in each note, then work the example)

Parallel lines: Same slope, different y-intercepts: _____ solution (a _____ statement).

Example 1

Solve.

2x + y = 4

4x + 2y = 3

Same line: Same slope and y-intercept: _____ solutions (a _____ statement).

Example 2

Solve.

$x - 3y = 6$

$-2x + 6y = -12$

Different slopes: Different slopes: _____ solution.

Example 3

Without solving, how many solutions does the system have?

$y = -x + 5$

$y = 3x + 5$

Example 4

Two streaming services: A charges $8 per month plus a $20 fee; B charges $8 per month plus a $15 fee. Will the costs ever be equal?

Part 3 · Partner practice

Partner 1

How many solutions? $y = 4x - 2$ and $8x - 2y = 4$

Partner 2

Solve: $3x - y = 2$ and $6x - 2y = 5$.